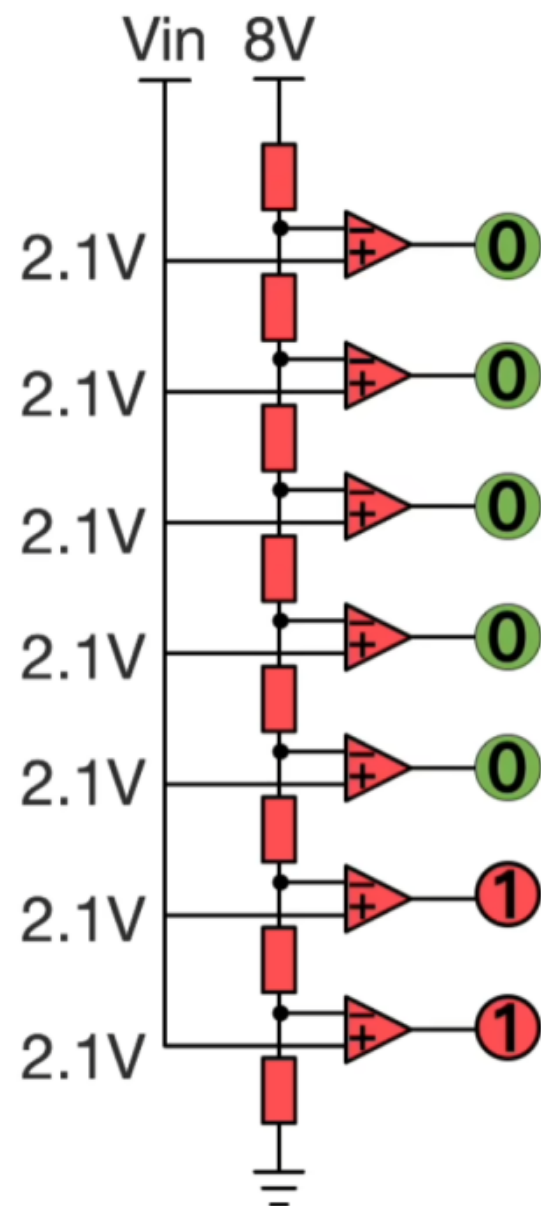
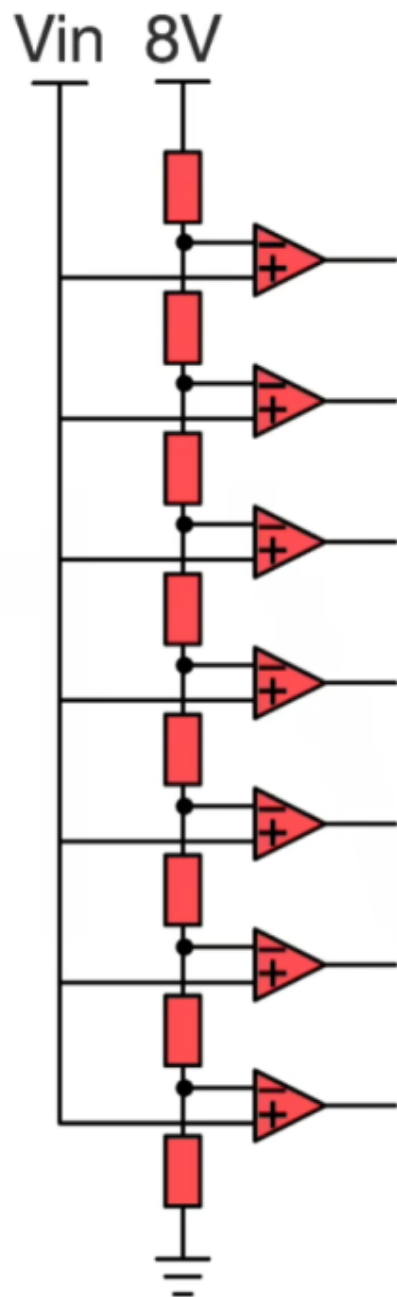
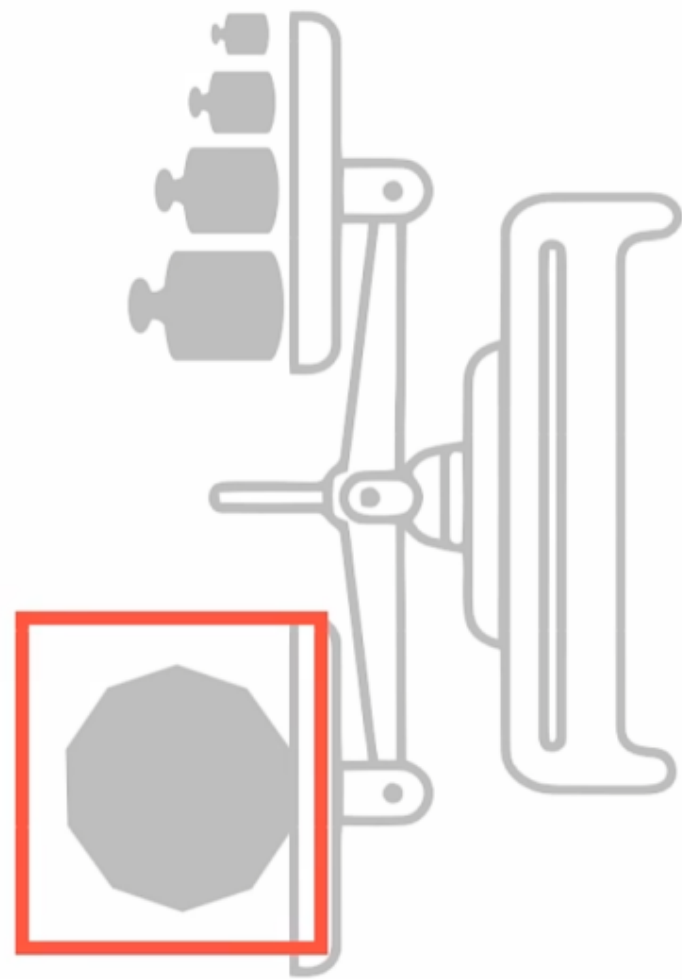
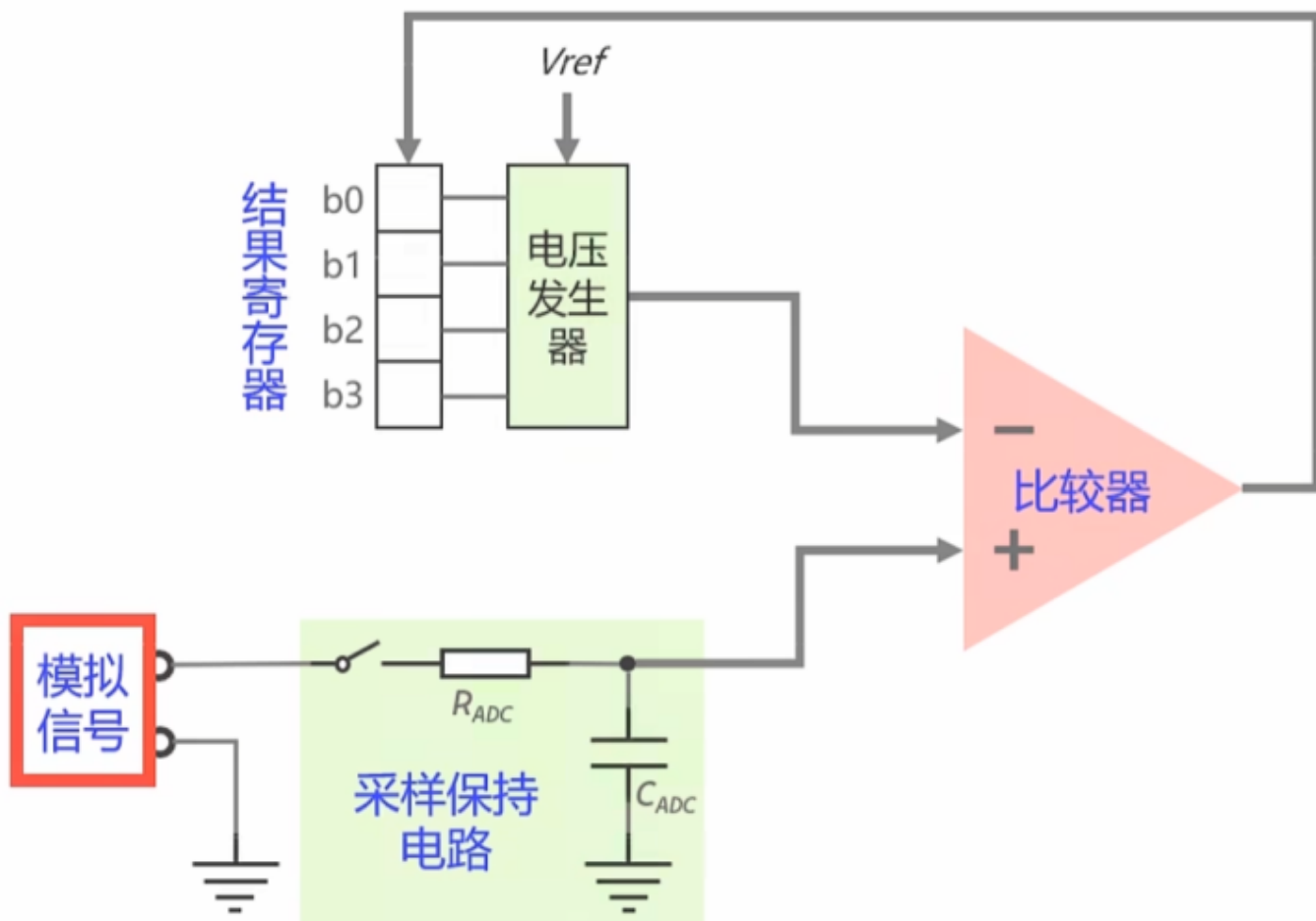


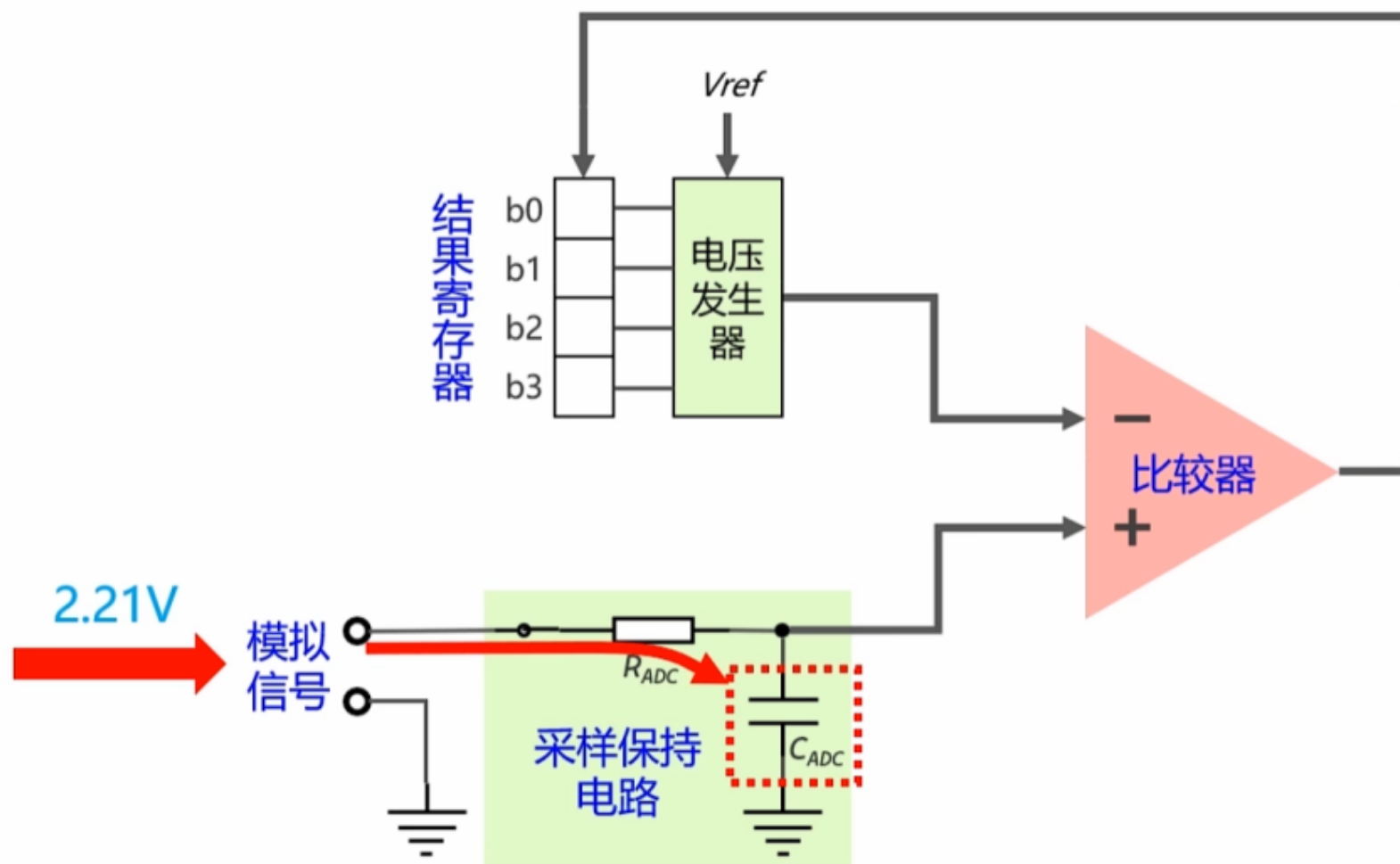


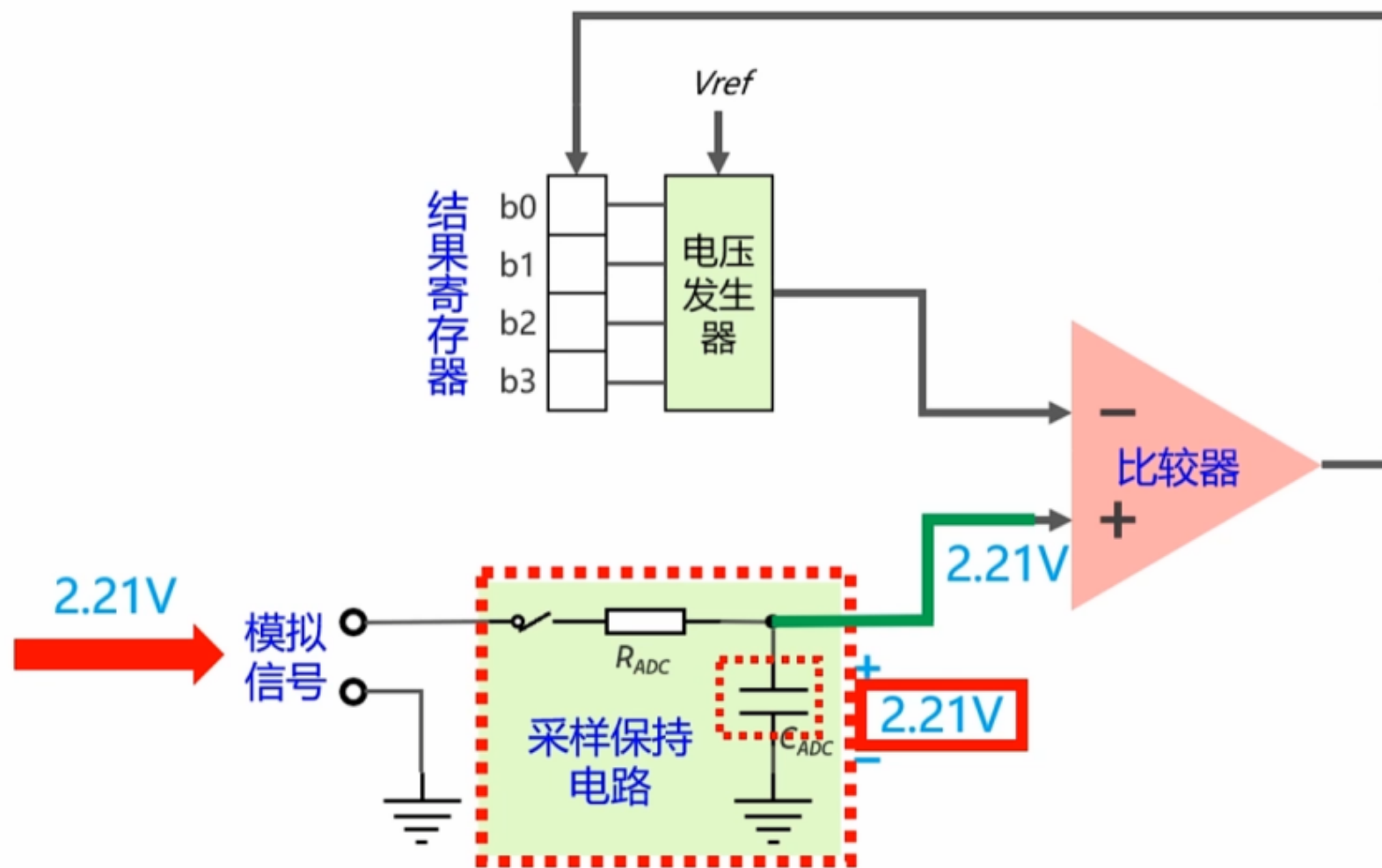
逐次逼近型ADC内部结构（采样深度=4bit）



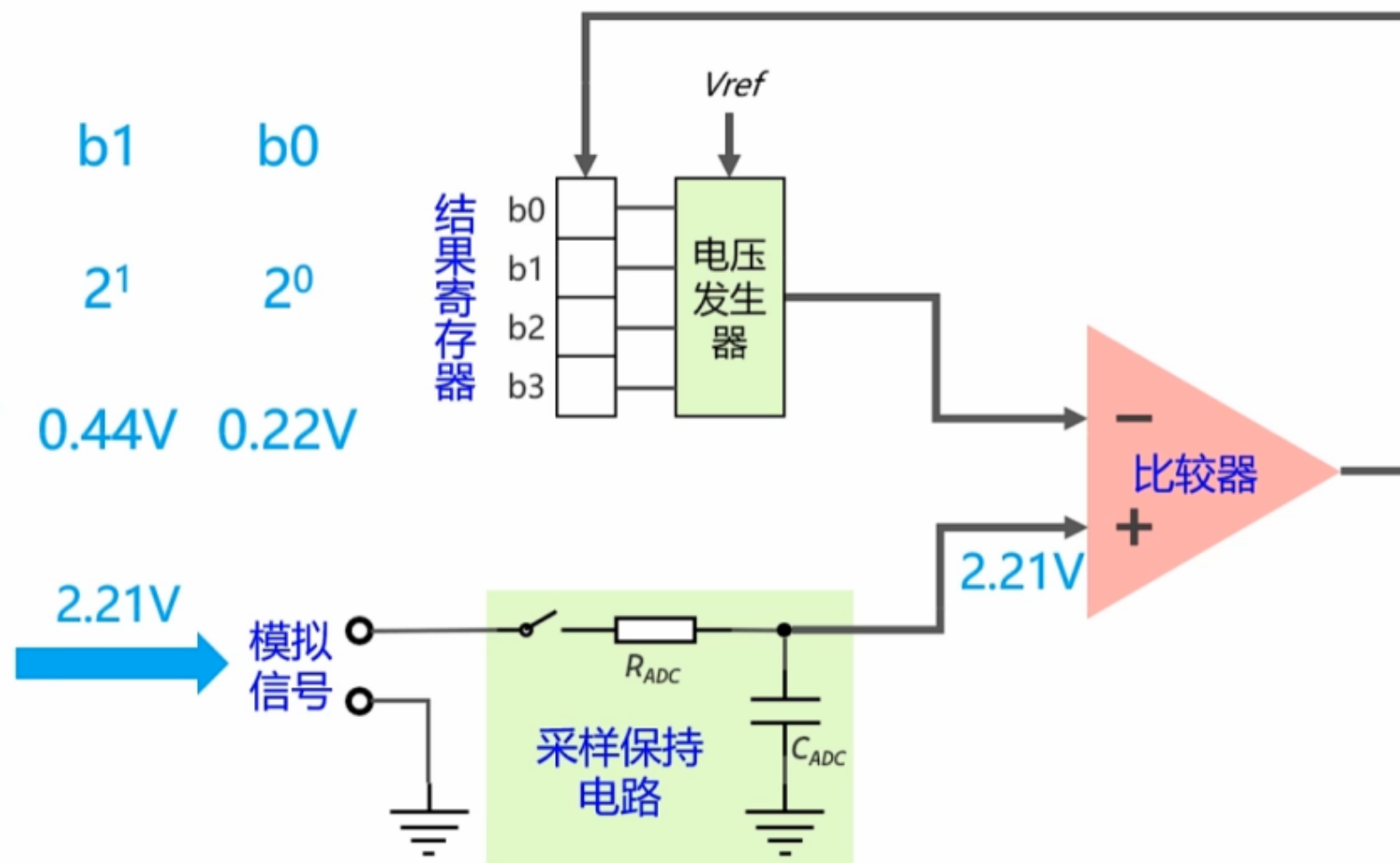


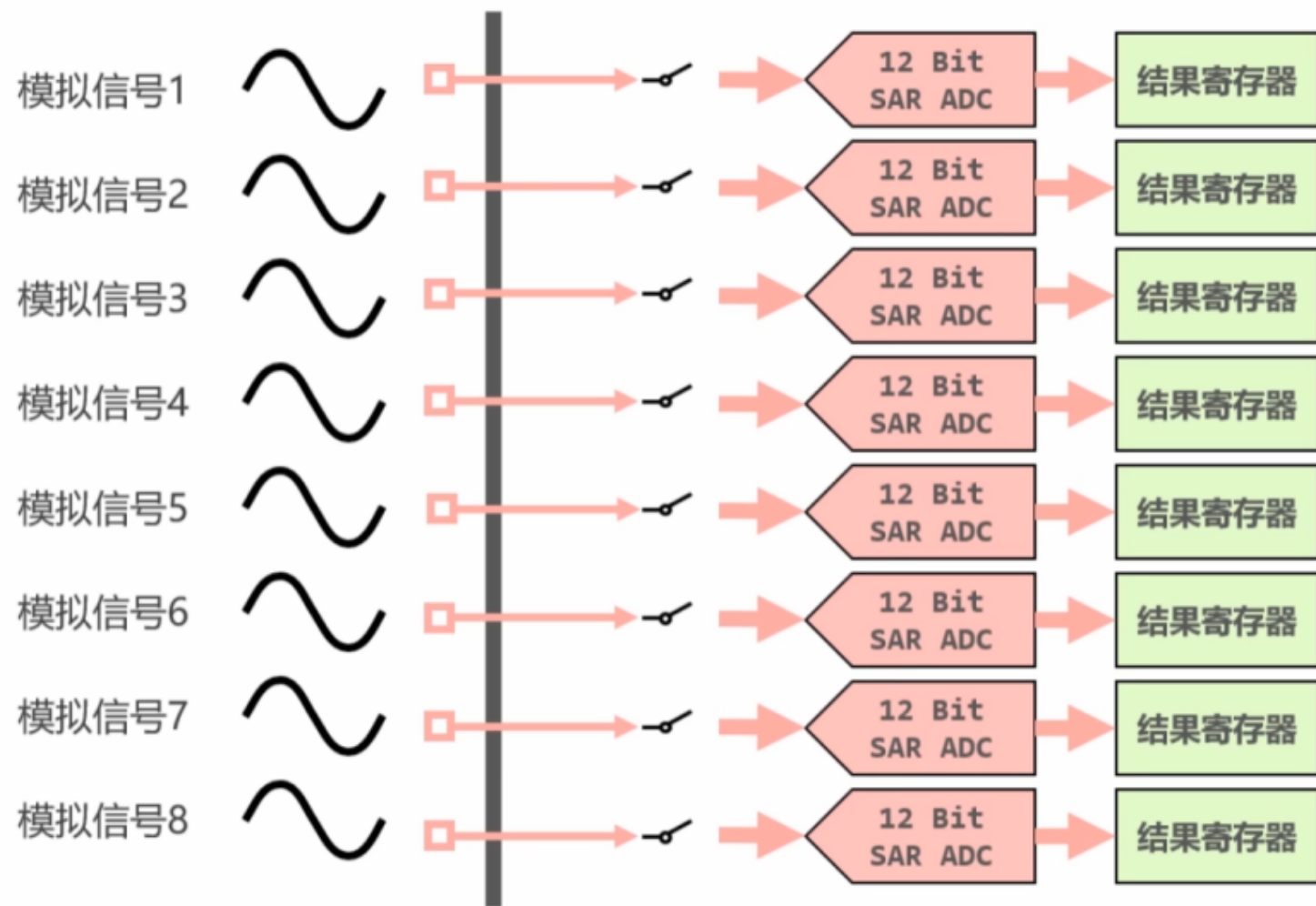
$$\text{每份} = \frac{(3.3 - 0)}{2^4 - 1} = 0.22V$$

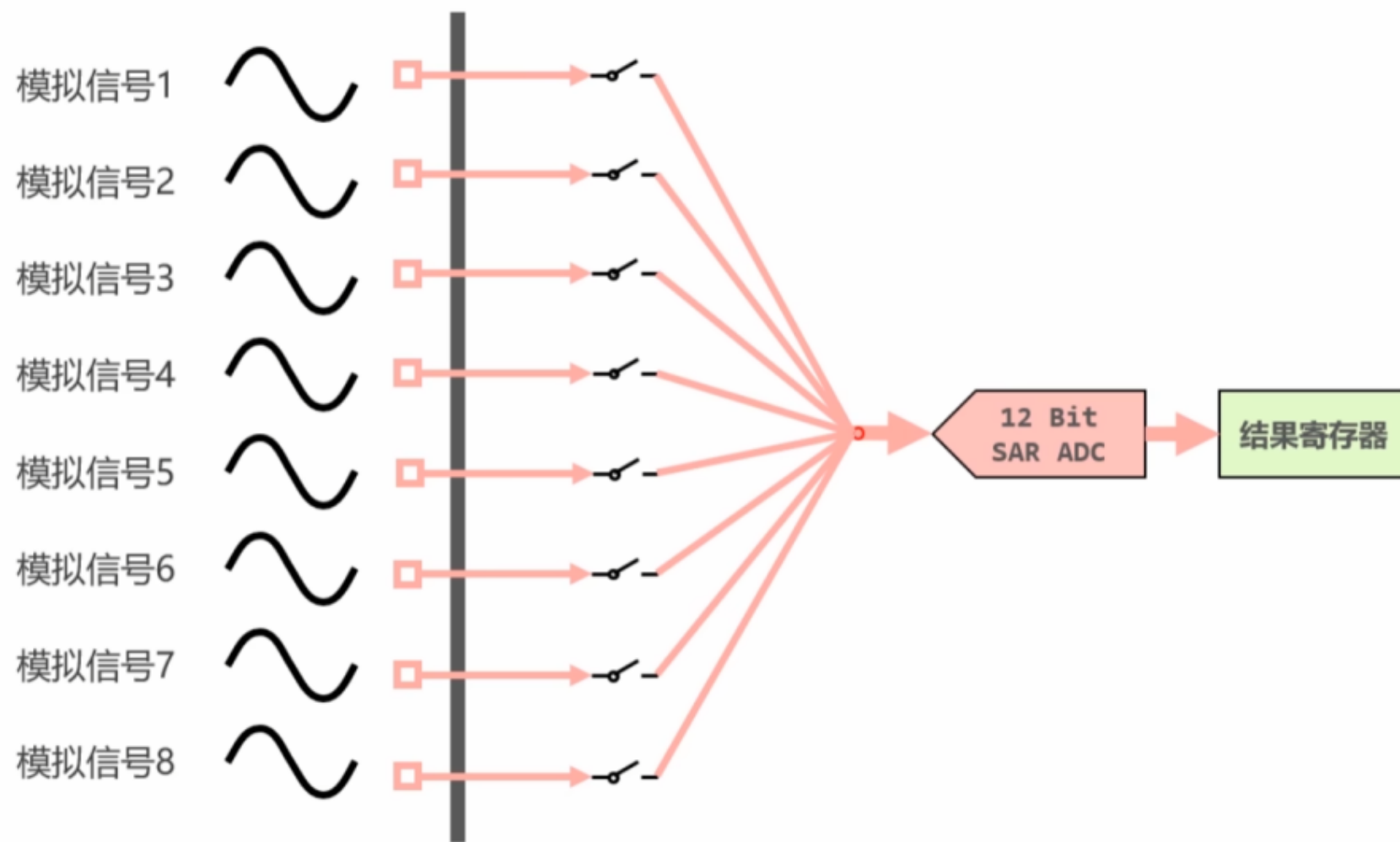


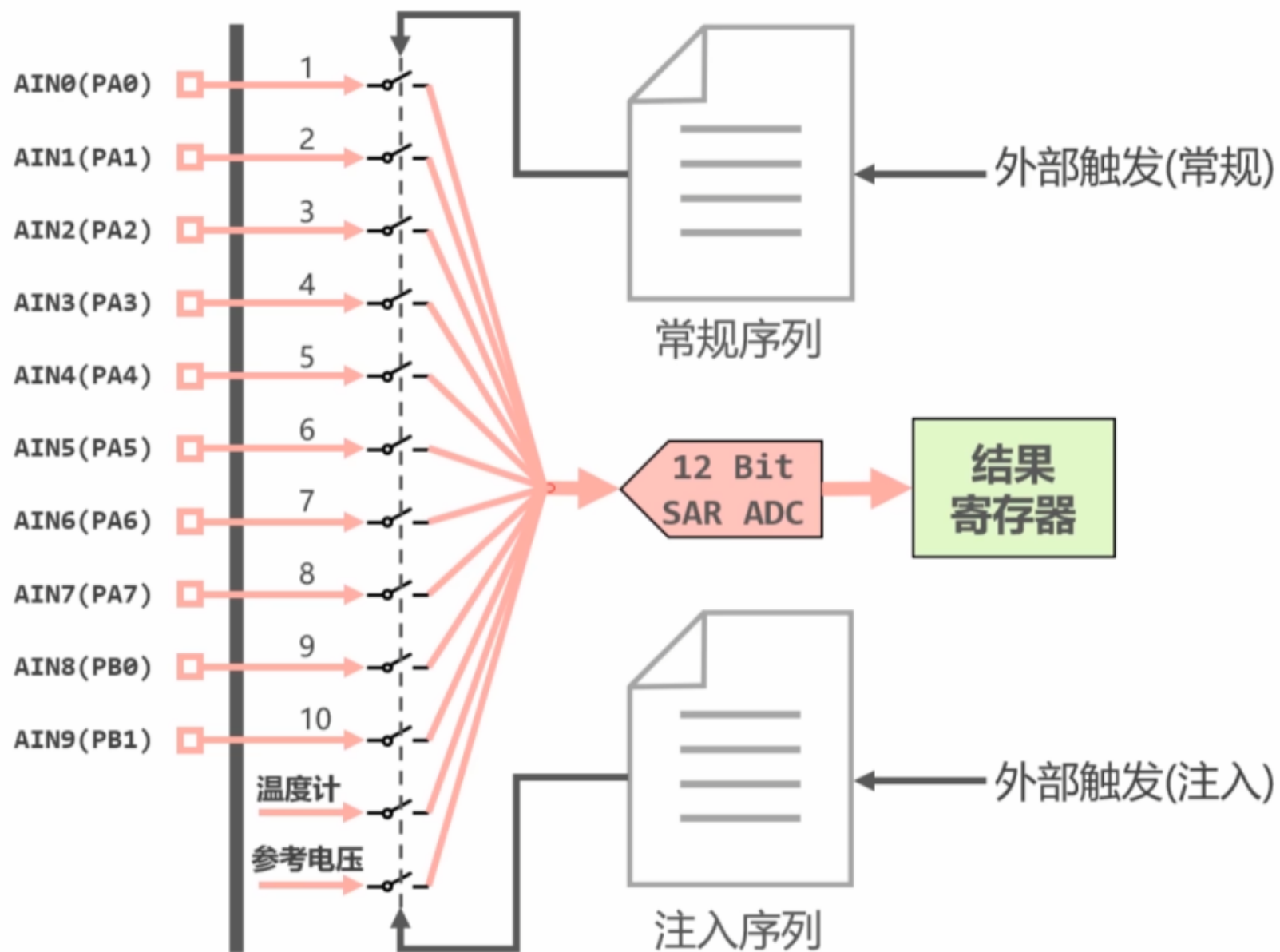


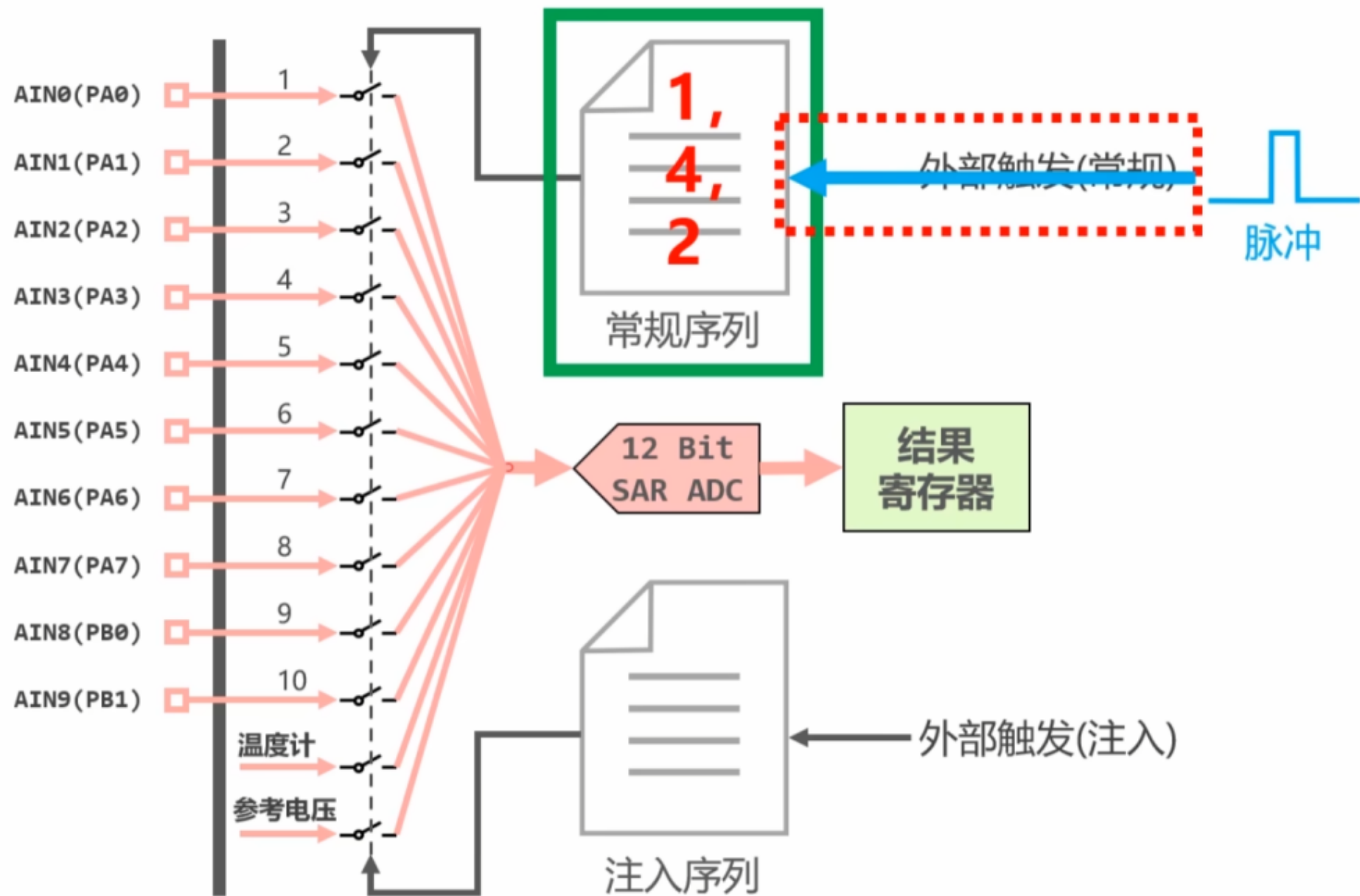
b3	b2	b1	b0
2^3	2^2	2^1	2^0
1.76V	0.88V	0.44V	0.22V

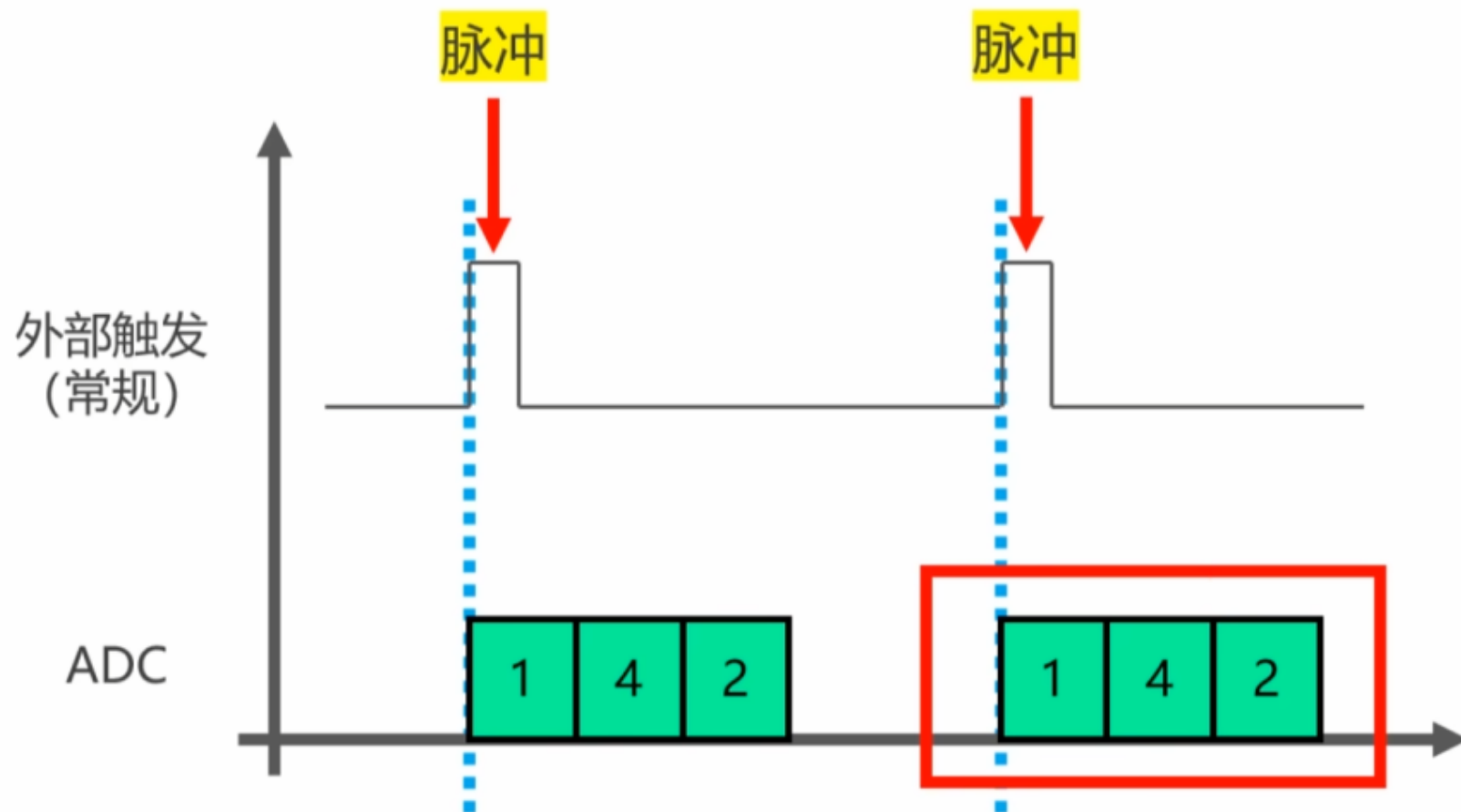












为什么要介绍ADC模块的时钟呢？

采样时间

转换时间

都需要表示成时钟周期的倍数



$$\text{时钟周期} = \frac{1}{14\text{MHz}} = \frac{1}{14} \mu\text{s}$$

~~采样时间 = 0.1us 转换时间 = 0.9us~~

$$\text{采样时间} = 0.1\mu\text{s} = 1.5 * 1/14\mu\text{s} = 1.5 \text{ cycle}$$

$$\text{转换时间} = 0.9\mu\text{s} = 12.5 * 1/14\mu\text{s} = 12.5 \text{ cycle}$$